

5. (15 pts.) Qual deve ser o valor de a para que o sistema

$$\begin{cases} x + 2y + z = 3 \\ 2x + y - z = 4 \\ x - 2y + az = 5 \end{cases}$$

seja impossível?

- (a) $a = 1$
- (b) $a = -3$
- (c) $a = 0$
- (d) $a = 3$
- (e) $a = -1$

$$x + 2y + z = 3$$

$$2x + y - z = 4$$

$$x - 2y + az = 5$$

Método de Gauss-Jordan:

$$C = \begin{bmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & -1 & 4 \\ 1 & -2 & a & 5 \end{bmatrix} \text{ op1: } L2 \rightarrow L2 - 2*L1$$

$$C = \begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & -3 & -3 & -2 \\ 1 & -2 & a & 5 \end{bmatrix} \text{ op2: } L3 \rightarrow L3 - L1$$

$$C = \begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & -3 & -3 & -2 \\ 0 & -4 & a-1 & 2 \end{bmatrix} \text{ op3: } L2 \rightarrow L2 * (-\frac{1}{3})$$

$$C = \begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 1 & 1 & 2/3 \\ 0 & -4 & a-1 & 2 \end{bmatrix} \text{ op4: } L1 \rightarrow L1 - 2*L2$$

$$C = \begin{bmatrix} 1 & 0 & -1 & 5/3 \\ 0 & 1 & 1 & 2/3 \\ 0 & -4 & a-1 & 2 \end{bmatrix} \text{ op5: } L3 \rightarrow L3 + 4*L2$$

$$3 - 2*(2/3) = 3 - 4/3 \rightarrow 5/3$$

$$C = \begin{bmatrix} 1 & 0 & -1 & 5/3 \\ 0 & 1 & 1 & 2/3 \\ 0 & 0 & a+3 & 14/3 \end{bmatrix} \text{ Se } a = -3, c_{34} = 0:$$

$$C = \begin{bmatrix} 1 & 0 & -1 & 5/3 \\ 0 & 1 & 1 & 2/3 \\ 0 & 0 & 0 & 14/3 \end{bmatrix}, \text{ para } a = -3. \text{ Feito :D}$$

4. (20 pts.) Calcule a inversa da matriz

$$A = \begin{bmatrix} -1 & 2 & -2 \\ -3 & 2 & -2 \\ -2 & 1 & 0 \end{bmatrix}.$$

Resp.:

$$A^{-1} = \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$$

$$A_3 = \begin{bmatrix} -1 & 2 & -2 \\ -3 & 2 & -2 \\ -2 & 1 & 0 \end{bmatrix}, I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$[A_3 | I_3] = \begin{bmatrix} -1 & 2 & -2 & 1 & 0 & 0 \\ -3 & 2 & -2 & 0 & 1 & 0 \\ -2 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \text{ op1: } L1 \rightarrow (-1)*L1$$

Realizar Gauss-Jordan:

$$[A_3 | I_3] = \begin{bmatrix} 1 & -2 & 2 & -1 & 0 & 0 \\ -3 & 2 & -2 & 0 & 1 & 0 \\ -2 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \text{ op2: } L2 \rightarrow L2 + 3*L1$$

$$[A_3 | I_3] = \begin{bmatrix} 1 & -2 & 2 & -1 & 0 & 0 \\ 0 & -4 & 4 & -3 & 1 & 0 \\ -2 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \text{ op3: } L3 \rightarrow L3 + 2*L1$$

$$[A_3 | I_3] = \begin{bmatrix} 1 & -2 & 2 & -1 & 0 & 0 \\ 0 & -4 & 4 & -3 & 1 & 0 \\ 0 & -3 & 4 & -2 & 0 & 1 \end{bmatrix} \text{ op4: } L2 \rightarrow L2*(-1/4)$$

$$[A_3 | I_3] = \begin{bmatrix} 1 & -2 & 2 & -1 & 0 & 0 \\ 0 & 1 & -1 & \frac{3}{4} & \frac{-1}{4} & 0 \\ 0 & -3 & 4 & -2 & 0 & 1 \end{bmatrix} \text{ op5: } L1 \rightarrow L1 + 2L2$$

$$[A_3 | I_3] = \begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} & \frac{-1}{2} & 0 \\ 0 & 1 & -1 & \frac{3}{4} & \frac{-1}{4} & 0 \\ 0 & -3 & 4 & -2 & 0 & 1 \end{bmatrix} \text{ op6: } L3 \rightarrow L3 + 3*L2$$

$$[A_3 \mid I_3] = \begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} & \frac{-1}{2} & 0 \\ 0 & 1 & -1 & \frac{3}{4} & \frac{-1}{4} & 0 \\ 0 & 0 & 1 & \frac{1}{4} & \frac{-3}{4} & 1 \end{bmatrix} \text{ op7: } L2 \rightarrow L2 + L3$$

$$[A_3 \mid I_3] = \begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} & \frac{-1}{2} & 0 \\ 0 & 1 & 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & \frac{1}{4} & \frac{-3}{4} & 1 \end{bmatrix} = [I_3 \mid A_3^{-1}]$$

$$A_3^{-1} = \begin{bmatrix} 1/2 & -1/2 & 0 \\ 1 & -1 & 1 \\ 1/4 & -3/4 & 1 \end{bmatrix}$$